

Advancing Armenian Inscription Recognition

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Abstract

Armenian monuments are rich in carved stone inscriptions. These inscriptions serve as vital records of cultural and linguistic heritage, offering insights into the lives, beliefs, and traditions of Armenians during the Middle ages. However, detecting and comprehending these inscriptions pose significant challenges. Due to weathering, vandalism, erosion, and the complexity of ancient scripts, many of these texts remain unreadable. Yet, the few existing studies indicate that deciphering these messages from the past is feasible with technological advancements. In the present project we study a unique, newly created and unexplored collection of digital twins of Armenian tapanakars (tombstones) and khachkars (cross-stones) focusing on hierarchical segmentation of the images using the detected geometrical and statistical features. The results are applied to character classification and the accuracy of the generated images is estimated. Since the detection stage of the algorithm is universal for any kind of shapes, it opens up new research avenues that extend beyond text recognition alone. The same pipeline can be adapted to identify decorative motifs, geometric symbols, and other visual patterns commonly found on tapanakar surfaces.

CCS Concepts

•**Computing methodologies** → **Computer vision problems**; Supervised learning by classification; •**Information systems** → **Optical character recognition**; Document representation;

1 Introduction

The motivation for this research arose during an effort to match names from tombstone inscriptions in a rural Armenian cemetery with records in a genealogical family tree. The slow and error-prone nature of manual transcription, particularly under conditions of surface degradation and stylistic variability highlights the need for computational assistance in the documentation of Armenian epigraphy.

The significance of this project lies in its potential to bridge the past and future by preserving and deciphering Armenian stone inscriptions, an invaluable yet often inaccessible historical resource. These inscriptions serve as vital records of cultural, religious, and linguistic heritage, offering insights into the lives, beliefs, and traditions of Armenians during the Middle ages. Due to weathering, vandalism, erosion, and the complexity of ancient scripts, many of these texts remain unreadable. By leveraging machine learning and 3D modeling, the project not only enhances inscription recognition and interpretation but also addresses the critical challenge of data scarcity. It supports the broader humanistic goal of safeguarding Armenian heritage. Armenian epigraphic inscriptions are deeply embedded in the cultural identity, linguistic development, and historical continuity of the Armenian people [TH24]. Their destruction or erasure constitutes a profound loss not only for Armenia but for global cultural heritage. Preserving and digitizing these inscrip-

tions under threat, especially in regions such as Artsakh, represents both an academic imperative and a form of cultural resilience in response to ongoing geopolitical pressures.

2 Related Works

The field of automatic recognition of Armenian inscriptions remain limited. To our knowledge, the first systematic paper was published only in late 2024 [VGDP24], although the documentation of the ancient inscriptions spans over almost 150 years, compare, for example [Ali90] and [Har23]. The most commonly used approach completes two stages the character detection and its subsequent classification [VG22]. The former has been tested for different alphabets, while the latter remains largely prospective, with data scarcity being the main challenge [VGDP24].

3 Dataset

Library of digital twins of Khachkars and Tapanakars Recent efforts by the 3DArmenia Initiative resulted in a new collection of digital twins of Armenian tombstones (tapanakars) and cross-stones (khachkars) [AS25]. Since many of these ancient stones have inscriptions, this collection of digital twins provides an unprecedented resource for historical and epigraphic research. The methodology to create high-quality 3D models involves a combination of LiDAR and photogrammetric scanning. LiDAR devices



Figure 1: 3D model overview

target a surface with a laser beam and measure the time for the reflected light to return to the receiver enabling it to create the exact spatial replica of the target. The equipment included a Faro M70 LiDAR scanner and a Sony A7RIII camera for photogrammetric scanning, with 1 to 2 LiDAR scanning positions per object and 60 to 300 photographs. LiDAR data was processed using Faro Scene and Leica Cyclone software, then combined with photometry data and processed into the final 3D model using RealityCapture software. The resulting model density is approximately 50100 million points. It is reduced to around 500,000 points to optimize for external hosting (see Figure 1).

Photo Collection A dataset of 100 inscription images was collected via fieldwork at historical cemeteries and church walls across Armenia, spanning the 17th–20th centuries. It includes several hundred lines of text and around 7,000 symbols, capturing diverse styles and materials. Compared to the 3D inscription dataset (which shows higher erosion), these samples exhibit low to medium surface damage, enabling more reliable symbol detection while still covering a range of degradation. The script is predominantly standardized *Erkatagir*, with less stylistic variation than the 3D set, making this dataset well-suited for testing preprocessing robustness and model generalization.

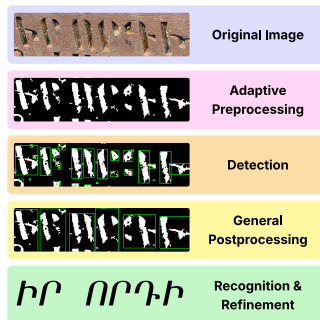


Figure 2: Sample Passthrough

4 Methodology

We propose a pipeline that integrates several components: adaptive preprocessing, image-based parameter regression, symbol region detection, and classification with refinement (See Figure 2). Each module is designed to handle the visual complexity of stone-carved texts, including degradation, background interference, and stylistic variability.

4.1 Adaptive Preprocessing

We develop a preprocessing configuration module that generalizes across diverse inscription images with varying wear, lighting, and texture. It predicts optimal parameters based on image-level metrics (see Figure 3).

A regression model is trained to map these metrics to preprocessing settings. At inference, no ground truth is needed, only the input image metrics: mean brightness, contrast, blur (Laplacian variance), edge density (Canny filter [Can86]), noise estimate, foreground ratio (Otsu method [LCC01]), entropy, dimensions, and aspect ratio. Predicted mask accuracy is measured via Dice and IoU [Huy22].

We evaluate three configurations: A (5 parameters), B (9), and C (16), progressively adding contrast, thresholding, morphology, and postprocessing to balance simplicity and performance.

4.2 Detection and Postprocessing

The system performs symbol detection using connected component analysis and geometric reasoning. The goal is to extract glyph-like structures from binarized patches and organize them according to the visual and spatial regularities present in carved inscription text. Overall, a two-stage detection approach consisting of DBSCAN-based component clustering [Sar21] followed by centroid-based K-Means line inference [CW23] proves effective in handling spatial variability, degradation, and irregular layouts, making it suitable for historical inscriptions that depart from modern typographic norms.

4.3 Alphabet-specific Features

The proposed pipeline is formulated and tested in an alphabet-independent form. However, the shapes of Armenian letters exhibit structural features that can be used in the text detection phase. We propose an algorithm based on a perception of Armenian words as well-structured patterns of vertical strokes, and apply it to grayscale images.

We extract vertical edges using the Sobel operator [Dav75], then invert and binarize them with a threshold of $T = 0$. To enhance edge continuity, we apply dilation and erosion. A median filter of radius 1 denoises the result. We then apply an FFT bandpass filter (filter_large = 40, filter_small = 30), followed by binarization. Centerlines of the binary regions are extracted [ZS84], and the Hough transform [IK88] is used to detect near-horizontal lines, each potentially representing a text line. Finally, we compute the horizontal projection of the vertical edge pattern to isolate individual text lines.

4.4 Recognition Model and Refinement

The classification module is currently implemented using a Support Vector Machine (SVM) classifier with Histogram of Oriented Gradients (HOG) as the feature representation. This combination is chosen for its strong performance in vision tasks with limited labeled data, as well as for its robustness to noise, scale changes, and minor geometric distortions properties that are critical in the context of historical inscriptions.

The refinement stage operates via a simple vocabulary-matching algorithm. If not all expected symbols within a patch are successfully identified, the system triggers a return to the detection stage with an alternative preprocessing configuration. This iterative loop

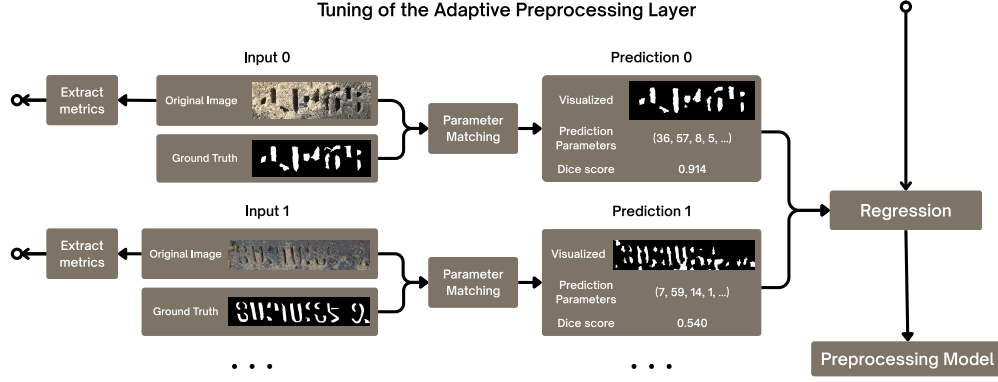
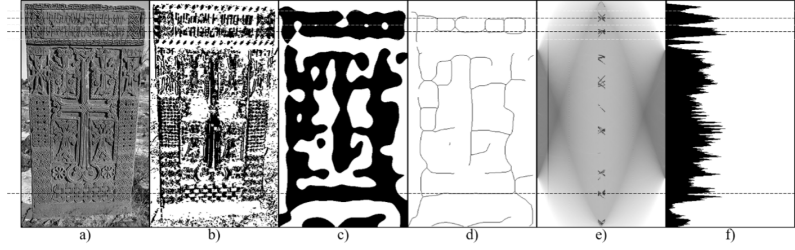


Figure 3

Figure 4: Text line detection steps: a) Original grayscale image; b) Enhanced denoised vertical edges; c) Binarized FFT filter removing fine details; d) Centerlines of binary regions; e) Hough accumulator near $\pi/2$ thresholded; f) Horizontal projection of (b).

is designed to improve recognition completeness and compensate for errors arising from local degradation or ambiguous shapes.

5 Results

5.1 Line Detection Stage

The text line detection algorithm was tested on a subset of kachkar images from the 3DArmenia Initiative dataset as a separate procedure. Figure 4 depicts the flow of the algorithm.

The top two black levels outline the detected individual text lines. Their heights are calculated based on the adjacent minimum points in the horizontal projection of the pattern of the vertical edges. The bottom level depicts a false positive that corresponds to the bottom layer of ornaments. The elimination of the false positives can be naturally incorporated in the Recognition Stage of the main pipeline. To further increase the precision, we propose to collect into a separate configuration all algorithm parameters and implement the adaptive preprocessing stage.

5.2 Symbol Detection Stage

We tested three preprocessing setups of increasing complexity. **Configuration A** applied basic filtering, thresholding, and morphology with 5 parameters. **Configuration B** added more blur and threshold types and allowed sequential morphological ops (9 parameters). **Configuration C** introduced heuristics and adaptive steps like edge fusion and contrast normalization, expanding to 16 parameters for greater flexibility on degraded inputs.

Table 1: Segmentation Prediction on Full-size patches

# Config	DoF ^a	IoU ^b Mean	IoU ^b Std	Dice Mean	Dice Std
A	5	0.48	0.17	0.63	0.15
B	9	0.60	0.15	0.74	0.12
C	16	0.64	0.13	0.77	0.10

^a Degrees of Freedom (i.e. Number of Parameters)

^b Intersection over Union

5.2.1 Evaluation on Full-size Patches

Configurations A, B, and C were evaluated on full inscription images, using genetic algorithms to optimize preprocessing per image. Performance was measured via mean and standard deviation of the Dice coefficient and Intersection over Union (IoU) across the test set. Dice scores near 80% and IoU values around 70% are considered near-optimal, given the ambiguity in pixel-level ground truth due to erosion, noise, and boundary uncertainty. The goal is accurate symbolic localization, not pixel-perfect replication. Overly high scores may suggest overfitting to noise rather than meaningful gains.

5.2.2 Evaluation on Grid-Split Regions

We evaluated **Configuration C** using uniform grid splits, where each tile was independently optimized via a genetic algorithm. This tile-wise strategy enables adaptive preprocessing tailored to local degradation, improving performance in spatially heterogeneous areas.

Grid sizes of 100, 75, and 50 pixels were tested with an 80/20 train-test split. As tile size decreased, Dice and IoU scores im-

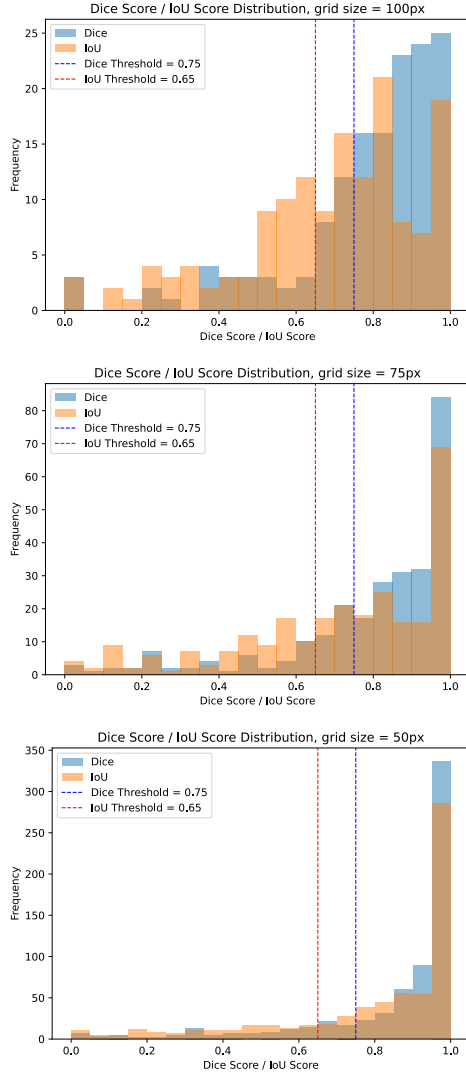


Figure 5: Dice / IoU Distributions per different grid sizes proved (see Figure 5). Table 2 summarizes acceptance rates (Dice ≥ 0.75 , IoU ≥ 0.65), average scores, and sample counts per grid size.

Table 2: Grid-wise performance comparison under fixed thresholds

Grid Size (px)	Accepted	Mean Dice	Mean IoU	Total Samples
100	62.2%	0.91	0.84	739
75	67.2%	0.93	0.87	1351
50	78.7%	0.96	0.92	3322

5.3 Recognition Stage

Noto Sans Armenian Classification results on a subset of confusing letterforms showed strong separability, even in degraded visual conditions. Early-stage evaluation of the prototype classifier focused on three visually similar classes, [ayb], [seh], and [veh], yielded a classification accuracy of approximately **85%**, validating the feasibility of extending the model to the full alphabet.

6 Conclusion & Future Studies

We presented a flexible pipeline for Armenian stone inscription analysis, combining classical computer vision and machine learning to segment glyphs under varied conditions. Our spatially-tuned detection strategy worked well even with limited data, supported by a proof-of-concept classifier. The approach generalizes beyond Armenian script, offering a foundation for scalable heritage OCR and visual motif detectionadvancing digital preservation of carved historical content.

Next steps involve integrating lightweight deep models (e.g., U-Net, Mask R-CNN, ViT) for improved handling of erosion and stylized scripts. We aim to develop an end-to-end trainable system combining segmentation, recognition, and context-aware correction via large language models. Fusing 3D features with 2D data may enhance detection under wear or poor lighting. Active learning, self-training, and synthetic data will expand and refine the training set. The system will also be extended to identify visual motifs and iconography for deeper epigraphic analysis.

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